

class12

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Section 4: Population Scale Analysis

Q13. Read this file into R and determine the sample size for each genotype and their corresponding median expression levels for each of these genotypes.

```
expr <- read.table("rs8067378_ENSG00000172057.6.txt", header=TRUE)
head(expr)
```

	sample	geno	exp
1	HG00367	A/G	28.96038
2	NA20768	A/G	20.24449
3	HG00361	A/A	31.32628
4	HG00135	A/A	34.11169
5	NA18870	G/G	18.25141
6	NA11993	A/A	32.89721

```
table(expr$geno)
```

A/A	A/G	G/G
108	233	121

```
by(expr$exp, expr$geno, median)
```

```
expr$geno: A/A  
[1] 31.24847
```

```
-----  
expr$geno: A/G  
[1] 25.06486
```

```
-----  
expr$geno: G/G  
[1] 20.07363
```

The dataset contains 462 total samples. Sample sizes for each genotype: A/A = 108, A/G = 233, G/G = 121. The median expression levels are: A/A = 31.25, A/G = 25.06, and G/G = 20.07. This shows a clear trend where having more G alleles is associated with lower ORM DL3 expression.

Q14. Generate a boxplot with a box per genotype, what could you infer from the relative expression value between A/A and G/G displayed in this plot? Does the SNP effect the expression of ORM DL3?

```
library(ggplot2)
```

Warning: package 'ggplot2' was built under R version 4.4.3

```
ggplot(expr, aes(x=geno, y=exp, fill=geno)) +  
  geom_boxplot(notch=TRUE) +  
  geom_jitter(alpha=0.3, width=0.2) +  
  labs(x="Genotype", y="Expression",  
        title="ORM DL3 Expression by Genotype") +  
  theme_bw() +  
  scale_fill_manual(values=c("red", "green", "blue")) +  
  theme(legend.position="none")
```

